1.1 RESISTOR MARKINGS

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Resistance value is marked on the resistor body. Most resistors have 4 bands. The first two bands provide the numbers for the resistance and the third band provides the number of zeros. The fourth band indicates the tolerance. Tolerance values of 5%, 2%, and 1% are most commonly available.

The following table shows the colors used to identify resistor values:

COLOR	DIGIT	MULTIPLIER	TOLERANCE	TC
Silver		x 0.01 W	±10%	
Gold		x 0.1 W	±5%	
Black	0	x 1 W		
Brown	1	x 10 W	±1%	$\pm 100*10^{-6}/K$
Red	2	x 100 W	±2%	$\pm 50*10^{-6}/K$
Orange	3	x 1 kW		$\pm 15*10^{-6}/K$
Yellow	4	x 10 kW		$\pm 25*10^{-6}/K$
Green	5	x 100 kW	±0.5%	
Blue	6	x 1 MW	±0.25%	$\pm 10*10^{-6}/K$
Violet	7	x 10 MW	±0.1%	$\pm 5*10^{-6}/K$
Grey	8	x 100 MW		
White	9	x 1 GW		$\pm 1*10^{-6}/K$

** TC – Temp. Coefficient, only for SMD devices

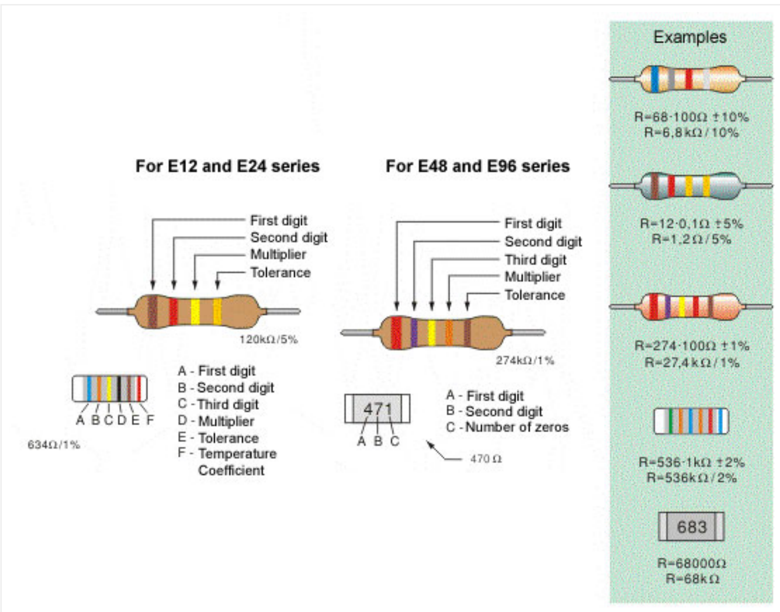
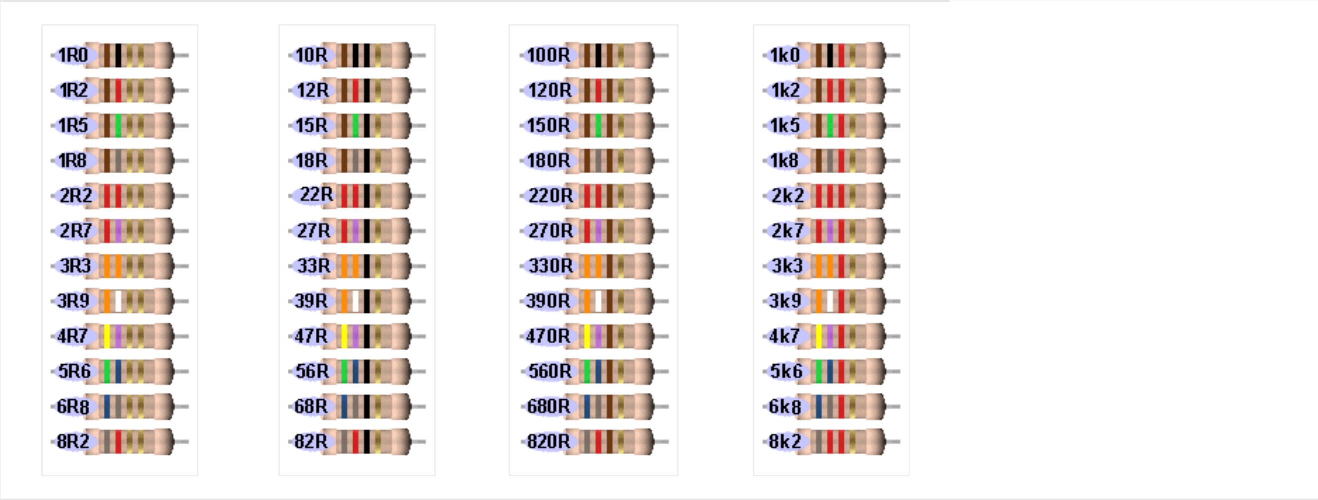
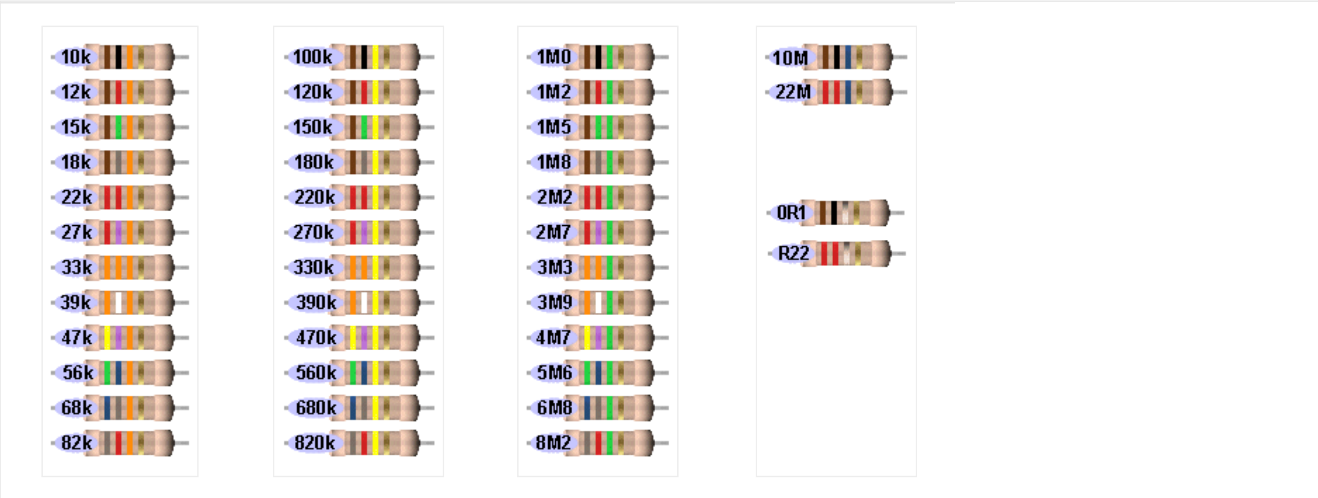


Fig. 1.2: b. Four-band resistor, c. Five-band resistor, d. Cylindrical SMD resistor, e. Flat SMD resistor

The following shows all resistors from 0R1 (one tenth of an ohm) to 22M:





NOTES:

The resistors above are “common value” 5% types.

The fourth band is called the “tolerance” band. Gold = 5%

(tolerance band Silver =10% but no modern resistors are 10%!!)

“common resistors” have values 10 ohms to 22M.

RESISTORS LESS THAN 10 OHMS

When the **third** band is gold, it indicates the value of the “colors” must be divided by 10.
Gold = “divide by 10” to get values 1R0 to 8R2
See 1st Column above for examples.

When the **third** band is silver, it indicates the value of the “colors” must be divided by 100.
(Remember: more letters in the word “silver” thus the divisor is “larger.”)
Silver = “divide by 100” to get values 0R1 (one tenth of an ohm) to 0R82
e.g: 0R1 = 0.1 ohm   0R22 = point 22 ohms
See 4th Column above for examples.

The letters “R, k and M” take the place of a decimal point. The letter “E” is also used to indicate the word “ohm.”
e.g: 1**R**0 = 1 ohm   2**R**2 = 2 point 2 ohms   22**R** = 22 ohms
2**k**2 = 2,200 ohms   100**k** = 100,000 ohms
2**M**2 = 2,200,000 ohms

Common resistors have 4 bands. These are shown above. First two bands indicate the first two digits of the resistance, third band is the multiplier (number of zeros that are to be added to the number derived from first two bands) and fourth represents the tolerance.

Marking the resistance with five bands is used for resistors with tolerance of 2%, 1% and other high-accuracy resistors. First three bands determine the first three digits, fourth is the multiplier and fifth represents the tolerance.

For SMD (Surface Mounted Device) the available space on the resistor is very small. 5% resistors use a 3 digit code, while 1% resistors use a 4 digit code.

Some SMD resistors are made in the shape of small cylinder while the most common type is flat. Cylindrical SMD resistors are marked with six bands – the first five are “read” as with common five-band resistors, while the sixth band determines the Temperature Coefficient (TC), which gives us a value of resistance change upon 1-degree temperature change.

The resistance of flat SMD resistors is marked with digits printed on their upper side. First two digits are the resistance value, while the third digit represents the number of zeros. For example, the printed number 683 stands for 68000W , that is 68k.

It is self-obvious that there is mass production of all types of resistors. Most commonly used are the resistors of the E12 series, and have a tolerance value of 5%. Common values for the first two digits are: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68 and 82.
The E24 series includes all the values above, as well as: 11, 13, 16, 20, 24, 30, 36, 43, 51, 62, 75 and 91. What do these numbers mean? It means that resistors with values for digits “39” are: 0.39W, 3.9W, 39W, 390W, 3.9kW,39kW, etc are manufactured. (0R39, 3R9, 39R, 390R, 3k9, 39k)

For some electrical circuits, the resistor tolerance is not important and it is not specified. In that case, resistors with 5% tolerance can be used. However, devices which require resistors to have a certain amount of accuracy, need a specified tolerance.